Take-Home Assignment

Assignment Goal: This project is designed to assess your capabilities in three key areas: data engineering, data analysis, and the use of GitHub for collaborative coding. You will be given an at-home healthcare scenario where you must process, analyze, and visualize data. Your solution, including both the final findings and the underlying code, must be submitted through a GitHub repository.

Scenario

LittleSteps, an at-home healthcare startup, aims to boost operational efficiency, particularly in nurse scheduling and patient visits. The company collects diverse data, but it often needs significant cleaning and processing for effective analysis. Your task is to support in analyzing this data to identify patterns in patient visit durations and nurse travel times. This analysis will be crucial for pinpointing and addressing areas for operational improvement.

Dataset Description

The following dataset is provided for analysis, detailing individual patient visits:

- [visits.csv](): This file contains visit-level information with the following attributes:
    - `visit_id`
    - `patient_id`
    - `nurse_id`
    - `visit_start_time`
    - `visit_end_time`
    - `service_type`
    - `visit_location`
    - `nurse_notes`

Submission Details & Deliverables

- **Collaborator:** Add [chanway-speedoc]() as a collaborator to your GitHub repository
- **Time Limit:** 4 calendar days from receipt of the assignment. Please inform us of any issues affecting this deadline
- **Repository Content:**
    - All Python scripts or Jupyter notebooks used
    - A comprehensive `README.md` file
    - Any generated plots or visual outputs

Tasks

Part 1: Data Preparation and Engineering

1. **Initial Data Exploration:**
    - Load the `visits.csv` file into a suitable data structure
    - Perform initial exploration to understand data structure, data types, and identify potential quality issues
2. **Data Cleaning & Preprocessing:**
    - **Missing Data:** Handle and provide justification for the treatment of missing values
    - **Date/Time:** Standardize all date/time formats
    - **Outliers:** Address outliers specifically in visit durations
    - **Duplicates:** Remove duplicate records based on `visit_id`
    - **Categorical Variables:** Clean and standardize categorical variables
    - **Text Data:** Extract relevant information from `nurse_notes` if possible

Part 2: Data Analysis and Visualization

1. **Feature Engineering**
    - Calculate and create two new columns: `visit_duration_minutes` and `travel_duration_minutes`
2. **Descriptive Statistics**
    - Calculate and present descriptive statistics for both `visit_duration_minutes` and `travel_duration_minutes`
3. **Core Analysis Questions**
    - Determine the average visit and travel duration across all service types
    - Analyze how average visit duration changes by `service_type`. Identify the service types with the longest and shortest average durations
    - Investigate if there is a significant difference in visit durations across various `visit_location` zones
    - Identify the top 3 and bottom 3 nurses by average travel duration to see who consistently has longer or shorter travel durations compared to the overall average
    - Infer insights from `nurse_notes` regarding visit outcomes or specific patient needs that may influence visit duration
    - Propose suggestions for improving operations efficiency based on your analysis
4. **Data Visualization**
    - Create clear and informative visualizations to support your findings. Examples include:
        - Histograms of `visit_duration_minutes`
        - Bar charts showing average visit duration by `service_type` and `visit_location`

        - Box plots illustrating the distribution of visit durations

Part 3: GitHub Competency and Documentation

1. **Repository Structure**
    - Ensure the GitHub repository is logically organized and easy to navigate (code, data, plots, etc.)
2. **`README.md` File**
    - The file must be comprehensive and located at the root of the repository, including:
        - A brief introduction to the assignment and your analytical approach
        - Instructions on how to set up the environment and execute your code
        - A summary of your key findings and insights, supported by your visualizations
        - Documentation of any assumptions made or challenges encountered
3. **Version Control**
    - Demonstrate good Git practices (e.g., clear commit history, meaningful commit messages)
4. **Code Quality**
    - Write clean, readable, and well-commented code
    - Utilize appropriate libraries
    - Structure your code into functions or logical blocks as necessary

Evaluation Criteria

Your submission will be assessed based on the following:

- **Correctness:** Accuracy of all calculations, data cleaning, and analysis
- **Completeness:** Successful completion and addressing of all outlined tasks
- **Code Quality:** Readability, maintainability, efficiency, and suitability of library usage
- **Analytical Rigor:** Depth of insights provided and clarity of interpretation of findings
- **Visualization Effectiveness:** Clarity, appropriateness, and impact of the visual representations
- **GitHub Proficiency:** Effectiveness of version control, logical repository structure, and comprehensive documentation